Advancing AR/VR: Innovative Strategies for Future Display Solutions

Sug Woo Jung, Sangho Park, Sang-Myeon HAN, JAE BEOM CHOI
Samsung Display Co. Ltd., Yongin-si, 17113, Republic of Korea

Abstract
OLED-on-Silicon (OLEDoS) has long served as the primary microdisplay platform for VR, yet recent advances are rapidly expanding its relevance to AR. This work presents a strategic and technology-driven reassessment of OLEDoS, highlighting innovations that overcome limitations of legacy WOLED-CF architectures. Key developments include fine-patterned RGB pixels enabled by sub-micron FSM alignment, leakage-suppressed pixel structures, ITO-based electrode engineering, and optical-cavity and microlens-assisted luminance enhancement. Together, these advances elevate RGB OLEDoS into a new performance class suitable for high-PPI, high-ACR AR/VR systems. By integrating device-level innovation with system-level requirements, we position OLEDoS not as an intermediate technology but as a strategic foundation for the next generation of immersive AR/VR displays.

Author Keywords
RGB OLEDoS; Sub-micron Pixel Patterning; Optical-Cavity Engineering; Microlens Array Extraction; High-PPI Microdisplay; AR/VR Displays.

1. Introduction
The rapid acceleration of augmented and virtual reality (AR/VR) technologies is reshaping the landscape of near-eye displays [1]. As form factors shrink while visual expectations continue to grow, microdisplays must simultaneously achieve extreme pixel densities, high luminance, wide color gamut, long life, and stable operation under demanding optical architectures such as pancake lenses and waveguides [1,2]. Classical boundaries between display physics, optical engineering, and system integration are dissolving into a unified design problem, one that must be solved holistically rather than at the level of a single component [2].

OLED-on-Silicon microdisplays have historically provided advantages such as high contrast and fast temporal response, particularly in folded lens systems such as pancake optics [2]. At the same time, microLED-on-Silicon (μLEDoS) has emerged as a strong candidate for AR due to its robustness, narrow spectrum, and extraordinarily high luminance, often exceeding one million nits at the device level [1]. This dichotomy, OLED for VR, microLED for AR, has shaped both academic narrative and industry strategies over the last several years [1]. However, this perspective fails to capture the pace of change within OLEDoS technology and overlooks a series of breakthroughs in pixel architecture, patterning fidelity, leakage control, optical extraction, and electrode engineering [3,8].

This work provides a renewed perspective on OLEDoS microdisplays for next-generation AR/VR systems. We examine limitations of legacy WOLED+CF implementations, identify key system-level requirements for AR and VR, and present a set of emerging innovations, validated through high-PPI RGB OLEDoS development, that collectively transform the potential of OLEDoS from a VR-centric technology into a scalable platform capable of significant AR penetration [3].

2. System Requirement for Next-Generation AR/VR Microdisplays
The modern AR/VR display manufacture imposes rigorous requirements on microdisplays that go far beyond traditional metrics such as resolution or peak luminance. For VR headsets, contrast ratio above 100,000:1, luminance over 10,000 nits, and wide color gamut are essential to support high dynamic range in dimly lit, enclosed optical paths. This means that pancakes and freeform lenses cause significant optical loss, so the luminance of the panel itself must be developed at least 3 to 5 times higher to meet the required luminance performance of the final product. [1,2].

AR displays impose an even more extreme demand profile. Waveguide-based combiners typically exhibit optical efficiencies of only 1–3%, meaning a display must produce up to 1,000,000 nits at the panel to achieve usable outdoor see-through performance [1]. In addition, AR systems require exceptionally stable angular color uniformity, narrow spectrum for minimizing waveguide crosstalk [1], and high ambient contrast ratio (ACR > 3:1 at 10,000 lux). These constraints collectively suggest that achieving AR-level performance requires more than incremental improvements to legacy OLED designs.

Nonetheless, the assumption that such levels are unattainable with OLED is outdated. Recent innovations in tandem stacking [7], microcavity engineering [9,10], dielectric mirrors, microlens arrays, optical collimation strategies, fine-pitch RGB patterning [3], and improved CMOS based driving schemes have increased the luminance and efficiency of OLEDoS devices far beyond their historical limits.

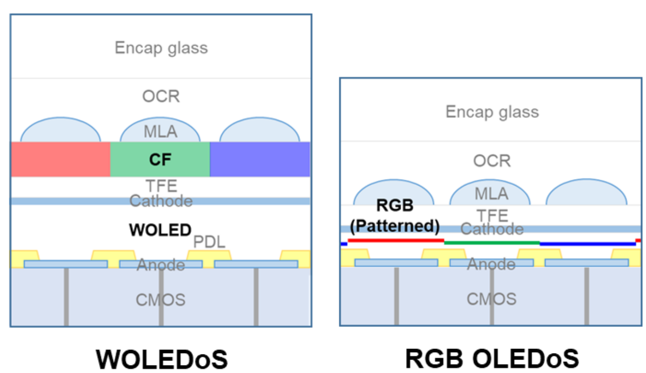

Figure 1. Structural comparison of WOLED–CF OLEDoS and RGB-patterned OLEDoS.

3. Limitations of Conventional OLEDoS and Drivers for Innovation
WOLED OLEDoS exhibits foundational limitations affecting luminance, spectral purity, and pixel scalability. The use of color filters introduces significant absorption, reducing luminance

ISSN 0097-996X/26/5701-0285-$1.00 © 2026

efficiency and increasing the required drive current [7], which accelerates thermal degradation [9]. Broad WOLED emission spectra poorly match optical systems requiring narrowband illumination, limiting compatibility with advanced optical engines and waveguides [1].

Fine Metal Mask (FMM) patterning, traditionally used for RGB OLED deposition on glass, becomes unstable at sub-micron apertures due to mechanical deformation and mask sagging, restricting pixel densities below approximately 700-1200 PPI. Additionally, at pixel pitches under ~3 µm, electrical and optical crosstalk degrade edge sharpness and color fidelity. These constraints motivate the transition toward directly patterned RGB OLED emitters on silicon.

These limitations motivate a new generation of OLEDoS engineering, a set of advances often not reflected in literature that still assumes outdated performance limits on OLED microdisplays.

4. Advances in RGB-Patterned OLEDoS

4.1 Sub-micron RGB patterning using Fine Silicon Masks

Recent developments demonstrate that OLEDoS can be dramatically more powerful when migrating from WOLED-CF to true RGB-patterned top-emission architectures. Fine Silicon Mask (FSM) technology provides sub-micron patterning fidelity without the deformation and alignment issues of FMMs. These masks enable pixel pitches below 2 µm, supporting displays with 5000–6000 PPI and above [2], [6]. This resolution levels align with the spatial sampling requirements of upcoming optical architectures. However, high PPI alone does not guarantee usable image quality. At these dimensions, leakage currents between adjacent subpixels, lateral optical crosstalk, and cavity-mode interactions significantly impact color purity and luminance uniformity.

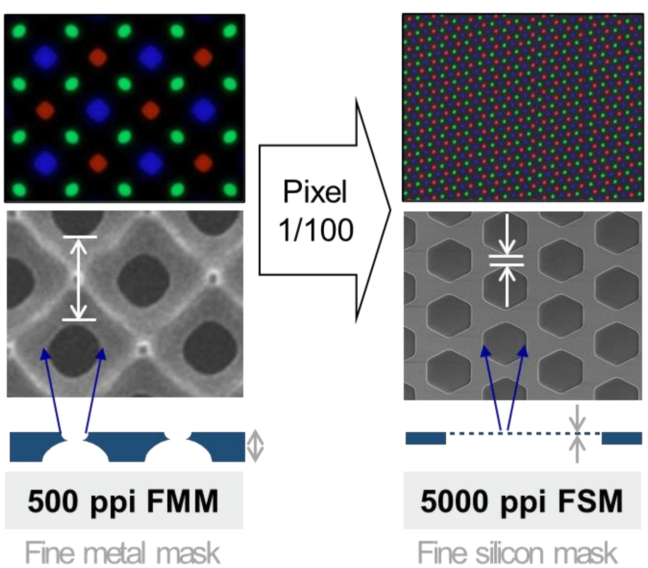

Figure 2. Pixel-perceptual comparison of RGB-patterned OLEDoS at different PPIs.

Innovative technologies, such as a leakage current-blocking PDL (Pixel Defined Layer) structure and a multilayer electrode combining mirror-like, highly reflective metal electrodes with high-work-function ITO, suppress electrical interference and limit parasitic light propagation. This allows each subpixel to maintain distinct emission, preserving color stability even under aggressive scaling.

Engineering the OLED microcavity allows emission spectra to be narrowed and directed, resulting in improved angular color uniformity and enhanced forward luminance—critical for compact lens systems and mixed-mode optical engines. Cavity tuning also improves wavefront consistency, which directly benefits see-through optical elements.

Microlens-based extraction redistributes captured optical modes into the forward emission path, typically resulting in a luminance enhancement of orders of magnitude, depending on the cavity and extraction geometry [9,10]. This optical gain reduces the electrical load on the OLED stack, improving reliability and thermal performance.

Collectively, these innovations constitute what may be termed RGB OLEDoS a significantly more advanced architecture than legacy WOLED-CF OLEDoS and one that is competitive with early µLEDoS prototypes for many AR/VR system requirements.

4.2 Perceptual image-quality enhancements

The transition from WOLED–CF architectures to RGB-patterned OLEDoS produces meaningful improvements in perceived image quality under practical viewing conditions. Figure 3 provides a direct comparison between the two architectures, illustrating the differences in luminance reproduction and low-gray-level fidelity that arise from their fundamentally different emission pathways.

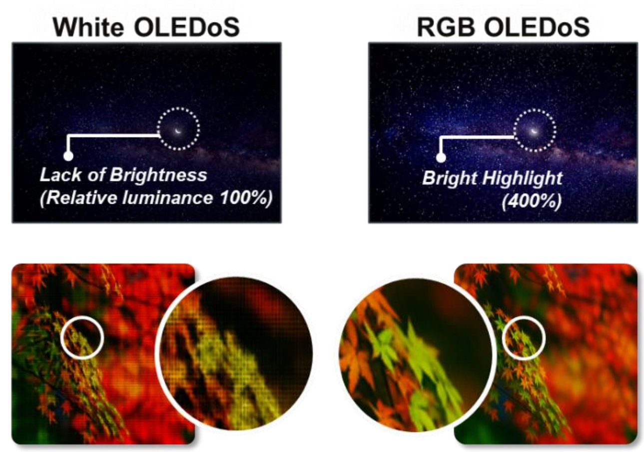

Figure 3. Luminance and spectral comparison between WOLED–CF and RGB OLEDoS.

Figure 3 highlights the practical visual impact of transitioning from WOLED–color-filter architectures to RGB-patterned OLEDoS. Due to the elimination of filter-related absorption and the use of cavity-enhanced RGB emitters, the RGB OLEDoS device achieves approximately four times higher luminance than the WOLED–CF panel under identical drive conditions. This luminance advantage directly translates into superior high dynamic range (HDR) performance, enabling brighter highlights, improved contrast in mixed scenes, and clearer reproduction of fine textures.

Equally important, the RGB device maintains clean separation of low-gray-level details, avoiding the shadow crushing and tone compression commonly observed in WOLED–CF microdisplays. As a result, dark-region visibility, subtle gradients, and fine depth cues remain intact, producing a more faithful and visually stable image. The combination of higher peak luminance and preserved low-level detail yields a display image that appears more vivid, sharper, and closer to the intended content representation, especially in scenes with both bright and dim elements.

Beyond perceptual benefits, these architectural improvements also translate into fundamental brightness and efficiency scaling advantages, as summarized in Section 4.3.

4.3 Performance scaling of RGB OLEDoS

WOLEDs have already been proven to approach 10,000 nits with an optimized stack architecture [11], and RGB OLEDoS without color filters can achieve stable luminance levels more than double this simply by improving the efficiency of the light-emitting materials. Furthermore, it is expected that these performance levels, when combined with cavity tuning and MLA extraction, can approach the luminance requirements of AR devices.

OLED's naturally low black level contributes to high ambient contrast ratios across diverse lighting environments. Furthermore, the tunability of OLED microcavities allows emission characteristics to be tailored to specific optical engines, enhancing compatibility and system-level efficiency. Collectively, these attributes underscore RGB-patterned OLEDoS as a practical, manufacturable microdisplay technology for future immersive systems.

4.4 AR-Focused Enhancements: Luminance, ACR, and Optical Matching

For AR devices, peak luminance is often cited as the primary limitation of OLEDoS. Yet luminance alone does not determine AR performance. What ultimately matters are ambient contrast ratio (ACR), angular color stability, and spectral compatibility with waveguides. By narrowing OLED spectra using microcavities, improving emission collimation using MLA/reflector hybrids, and tuning the half-power viewing angle, OLEDoS can achieve significantly higher optical coupling efficiency into waveguides than broad-spectrum WOLED sources. Even without achieving 1 million nit luminance, achieving a high ACR with controlled spectral shape can make OLEDoS viable for indoor and mixed-light AR environments.

AR-oriented OLEDoS, when combined with tandem stacks and low-voltage charge transport layers, can achieve luminance approaching tens of thousands of nits, which is sufficient for many waveguide and birdbath systems.

Therefore, OLEDoS is not a technology that is fundamentally excluded from augmented reality, but is gradually establishing itself as a practical technology as a display engine for next-generation mixed reality devices.

5. Conclusion

Advances in RGB-patterned OLEDoS, including sub-micron patterning, improved pixel isolation, optimized electrodes, and microcavity/MLA-based luminance enhancement, demonstrate that OLED-on-Silicon can overcome the long-standing inefficiencies of WOLED–CF architectures. These developments significantly improve luminance efficiency, spectral control, and pixel scalability, enabling performance levels suitable for modern high-PPI near-eye systems.

By enhancing luminance without excessive electrical load and delivering stable color and contrast across diverse optical engines, RGB OLEDoS now meets many requirements traditionally thought achievable only by microLED. Just as importantly, these improvements are realized within mature OLED and CMOS manufacturing infrastructures, offering a practical and scalable path for deployment.

Thus, RGB-patterned OLEDoS should be viewed not as a transitional technology but as a robust, manufacturable, and strategically relevant platform capable of powering future AR/VR displays.

6. Acknowledgements

The authors express their sincere appreciation to eMagin, an affiliated subsidiary of our organization, for its valuable technical support and collaboration throughout the development of this work.

7. References

[1] I. Sim, et al., Microdisplay technologies in augmented reality and virtual reality headsets, Nature Rev. Elec. Eng., 2, 634 (2025)

[2] J. Ha, et al., Advanced VR and AR displays: improving the user experience, Inform. Display, 39(3), 20 (2023)

[3] A. Ghosh, et al., Directly patterned 2645 PPI full color OLED microdisplay for head mounted wearables, SID Symposium Digest of Technical Papers, 47, 837 (2016)

[4] Y. Sun, et al., Enhanced light out-coupling of organic light-emitting devices using embedded low-index grids, Nature Photonics, 2, 483 (2008)

[5] S. Möller, et al., Improved light out-coupling in organic light-emitting diodes employing ordered microlens arrays, J. Appl. Phys., 91, 3324 (2002)

[6] J. Lee, et al., Hot excited state management for long-lived blue phosphorescent OLEDs, Nature Comm., 8, 15566 (2017)

[7] H. Cho, et al., White OLED microdisplay with a tandem structure, J. Information Display, 20, 249 (2019)

[8] A. Ghosh, et al., Ultra-high-brightness 2K x 2K full-color OLED microdisplay using direct patterning of OLED emitters, SID Symposium Digest of Technical Papers, 48, 226 (2017)

[9] N. C. Giebink, et al., Intrinsic luminance loss in phosphorescent OLEDs due to bimolecular annihilation reactions, J. Appl. Phys., 103, 044509 (2008)

[10] J. Kim, et al., Systematic control of the orientation of organic phosphorescent complexes for improved light outcoupling, Advanced Materials, 31, 1900921 (2019)

[11] J. Jo, et al., High-Luminance, Large-Size 4K OLED Microdisplays for VR/MR Applications, SID Sympo. Digest, 55 (1), 227 (2024)